

CSCE 676 DATA MINING · MODULE 12 · LECTURE NOTES

Privacy

What mining does to the people in the data.

MODULE OUTCOMES

- Explain why anonymisation fails.
- Explain k-anonymity and its limits.
- Explain differential privacy and what it guarantees.
- Explain the inference problems beyond identification.
- Assess a mining study's privacy properties.

1 Why anonymisation fails

REMOVING IDENTIFIERS DOES NOT ANONYMISE DATA, BECAUSE THE REMAINING ATTRIBUTES IDENTIFY PEOPLE

A handful of apparently innocuous attributes is usually unique. Date of birth, postcode, and sex together identify a large majority of a population — and none of the three is an identifier in the sense that any removal policy would recognise.

And the re-identifications are numerous and varied rather than exotic: a released hospital discharge dataset linked against a public voter roll; anonymised movie ratings linked against public reviews elsewhere; mobility traces identified from four visited locations. Each used only public auxiliary data.

The mechanism is always the same: link the 'anonymous' records against an auxiliary dataset that shares some attributes — and you cannot know what auxiliary data exists, or what will exist next year.

So 'anonymised' is not a property you can establish by inspecting your own dataset, which is what makes this a structural failure rather than a series of mistakes by careless people — and is the argument that motivated §3's definition.

AND HASHING AN IDENTIFIER DOES NOT ANONYMISE IT

A hashed email address, phone number, or national identifier is recoverable by enumeration — the input space is small enough to hash exhaustively on commodity hardware — which is **CSCE 711 Module 05 §4's third misuse**, arriving here with a privacy consequence.

And a hash is a *join key*: two parties who hash the same identifier with the same function can link their datasets without ever exchanging the identifier — which is **frequently the actual purpose of hashing it**, and is rarely the stated one.

So 'we only store hashed identifiers' is a claim about storage format rather than about privacy — and **it is presented to users and to regulators as though it were the latter**, which is CSCE 638 Module 12's bias-term conflation in a different register.

A salted, slow construction does change the answer (CSCE 711 Module 09) — but then it is no longer usable as a *join key*, because the salt is per-record, **which is usually the reason it is not used**.

2 k-anonymity

k-ANONYMITY

every record is indistinguishable from at least $k-1$ others on the quasi-identifiers. Achieved by generalising (exact age $\rightarrow$ age range) and by suppressing records that cannot be grouped.

THE HOMOGENEITY ATTACK

if all k records in a group share the same sensitive value, you learn that value without identifying anyone. k -anonymity says nothing at all about the sensitive attribute.

l-DIVERSITY patches that: require l distinct sensitive values within each group.

THE SKEWNESS ATTACK

l distinct values with one of them at 95% still leaks a great deal.

t-CLOSENESS patches that: require each group's distribution of the sensitive attribute to resemble the distribution in the whole dataset.

AND THE PATTERN IS THE PROBLEM: each definition is broken by an attack the previous one did not model, because none of them quantifies over all possible auxiliary information.

The patch sequence is the lesson rather than the definitions — a privacy definition that enumerates attacks is always one attack behind, because the next auxiliary dataset or the next correlation was not in the model. **The distinction between enumerating attacks and quantifying over them is what motivates**

§3, and it is a general lesson that applies well beyond privacy — it is the same argument as CSCE 711 Module 01 §3's definitional approach to security.

3 Differential privacy

$$P[M(D) \in S] \leq e^\epsilon \cdot P[M(D') \in S]$$

- For *all* datasets D and D' differing in a single record, and *all* sets of outputs S . The two universal quantifiers are where the strength lives.
- So your presence in the dataset barely changes the probability of any output — which bounds what any observer can learn about you specifically, whatever else they already know, because the bound holds against every possible auxiliary dataset by construction.
- ϵ is the privacy budget, and it composes: each query spends from it, and k queries at ϵ each gives a total of $k\epsilon$ — which is the practical constraint and the thing that makes deployment hard (§3's callout).
- The quantifier over *all* auxiliary information is what distinguishes this from §2's definitions — it is a property of the mechanism rather than of the data, which means it can be established by analysing the algorithm and does not depend on what else exists in the world. **That shift is the whole contribution.**

AND THE HONEST ACCOUNT OF WHAT IT COSTS

What it gives: a guarantee that holds against any adversary with any auxiliary information whatsoever, that composes predictably across queries, and that is a property of the mechanism rather than of the dataset — three things no earlier definition provided.

What it costs: accuracy, which degrades as ϵ falls — and critically, the loss is worst exactly where the data is sparse, which means small subgroups are harmed most by the noise, and that is an equity consequence worth stating (CSCE 638 Module 12 §2's disaggregation argument).

And the budget is the real operational difficulty. Every query spends from it, so an exploratory interactive analysis exhausts it quickly — which is why real deployments release a fixed, pre-decided set of statistics rather than answering arbitrary questions, and why the technology fits a census better than a data science team.

So the honest claim names ϵ and the composition — ' ϵ -differentially private' without the value of ϵ and without the number of queries is not a claim at all, and a large ϵ guarantees very little, which is worth checking when the term appears in a product description.

4 Beyond identification

- **Group inference.** A model trained on willing participants can predict a sensitive attribute about people who were never in the training data at all — which no per-record guarantee prevents, because the harm is statistical and operates through population-level correlation.
- **Inference of what was never collected** — pregnancy, health status, sexuality, political view, and immigration status are all predictable to some degree from ordinary behavioural data, **which makes 'we never asked' irrelevant as a privacy claim.**
- **Reconstruction from aggregates.** Enough published exact statistics determine the individual records — which is a linear algebra result rather than a leak or a bug, and is the one that surprises people:

publishing many exact aggregates is mathematically equivalent to publishing the microdata, which is why one national census adopted differential privacy.

- **And repurposing — data collected for one stated purpose used for another**, which is a **governance failure that no mathematical mechanism addresses** and is probably the most common actual harm.
- **So the question is not only 'can anyone be identified' but 'what can be inferred, about whom, and with what consequence'** — and the second question is the one a privacy assessment usually omits.

WHAT TO DO IN A MINING STUDY

Collect less. The data you do not hold cannot be breached, subpoenaed, repurposed, or re-identified — which is **CSCE 701 Module 01's reduce-the-surface principle** arriving as a privacy measure, and is the only one that is unconditionally effective.

Aggregate early and keep the raw data separate, with access logged and reviewed (CSCE 701 Module 09 §1's secret-access events) — so that the surface requiring the strongest protection is as small as possible.

Use differential privacy for anything published, with ϵ and the query count stated — and **do not claim anonymisation you have not established**, which §1 argues you generally cannot.

And state what can be inferred from your output — which is §4's question and is the genuinely hard part of a privacy assessment, since it requires thinking about what a determined party could do with what you released rather than about what you intended. **Project 2 requires this when the data concerns people.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Dwork & Roth — The Algorithmic Foundations of Differential Privacy (free PDF)	§3's first three chapters — the definition, the basic mechanisms, and composition, which is all this course needs.
Narayanan & Shmatikov — Robust De-anonymization of Large Sparse Datasets (free)	§1's mechanism, demonstrated — and the clearest account of why auxiliary information is the problem.
Machanavajjhala et al., and Li et al. — l-diversity and t-closeness (free)	§2's patch sequence in the originals — read them as a sequence, which is the lesson.
Dinur & Nissim — Revealing Information while Preserving Privacy (free)	§4's reconstruction result — the theorem that aggregates determine the records.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Compute how many people in a public census share your birth date, postcode, and sex.**
- 2 **Find two public datasets with overlapping attributes** and describe how they could be linked.
- 3 **Enumerate a hashed low-entropy space** — hash 10,000 plausible phone numbers and match them.
- 4 **Make a dataset 5-anonymous** by generalising, and measure the information you destroyed.
- 5 **Construct a homogeneity attack** against your own k-anonymous release.
- 6 **Implement the Laplace mechanism** for a count query and verify the noise scale.
- 7 **Measure the accuracy loss** against ϵ , and separately for a small subgroup.
- 8 **Compute the total budget** for a ten-query analysis.
- 9 **Reconstruct individual records** from a set of exact aggregates you publish.
- 10 **Write the privacy assessment** for Project 2, including what can be inferred from your output.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does removing identifiers not anonymise data?
- 2 What is the common mechanism in every re-identification?
- 3 Why can anonymisation not be established from your own dataset?
- 4 Why does hashing not anonymise, and what is a hash really?
- 5 Define k-anonymity and give the attack against it.
- 6 Give the patch sequence and say what the pattern shows.
- 7 State differential privacy and say what the quantifiers buy.
- 8 What does it cost, and who is harmed most?
- 9 Why is the budget the operational difficulty?
- 10 Name four harms that privacy mechanisms do not address.